Homework: classical ML baseline for skin-lesion classification (MILK10k)

Sebastián Ríos Saucedo · Computer Vision & Speech Recognition · EADA · 23 October 2026

Repository	https://github.com/sebarios02/Milk10k-project
Full report	reports/milestone2_classical_baseline.md
Feature extraction	milk10k_pipeline/features.py · scripts/09_extract_features.py
Feature sets, models, metrics	milk10k_pipeline/classical.py
Model selection	scripts/10_model_selection.py · outputs/classical/model_selection.csv · selected.json
Final evaluation (test)	scripts/11_final_evaluation.py · test_metrics_ci.csv · test_metrics.csv · test_confusion.csv
8-class stretch	scripts/12_fine_target.py · fine_metrics.csv
Run everything / tests	scripts/run_milestone2.py · tests/test_features.py
All deliverables	SUBMISSION.md

1. What was done

- **Task and data.** Benign / Indeterminate / Malignant (`diagnosis_1`) on the Milestone 1 **lesion-level splits**: 3,742 / 749 / 749 lesions, no lesion shared between splits. One sample = one lesion. The features of its dermoscopic and clinical images are concatenated. Leaky columns are never used.
- **Hand-crafted features from Sessions 2-4** (149 per image). **Colour histograms** (RGB, HSV). **Lesion-vs-skin colour statistics** in RGB, HSV and CIELab (contrast, ΔE, hue, colour variegation). **Edge/texture filters**: Sobel after a Gaussian, |LoG| at σ = 1, 2, 4, 9×9 local std, my Canny thresholds, entropy. **Shape** of an Otsu lesion mask (compactness, solidity, asymmetry, i.e. ABCD-inspired). The mask is colour-aware (darker + redder than the skin), because grayscale Otsu missed pale pink BCCs in the Session 4 homework. Plus age, sex and body site.
- **8 feature sets**: each family alone (F1-F4, dermoscopic), all dermoscopic (F5), both views (F6), both views + metadata (F7), metadata only (F0, reference). **5 models**: Logistic Regression, kNN, RBF-SVM, Random Forest, HistGradientBoosting, plus majority / stratified dummies. All are pipelines (imputer → StandardScaler → model) with `class_weight="balanced"` (Session 3).
- **Protocol.** Hyper-parameters by 5-fold stratified CV on **train**. The best (feature set, model) of 40 is chosen on **val** by **balanced accuracy** (plain accuracy is misleading: the majority dummy gets 69 %). The winner is refitted on train+val and scored **once on test**, with 1,000-resample bootstrap 95 % CIs.

2. Model selection (validation)

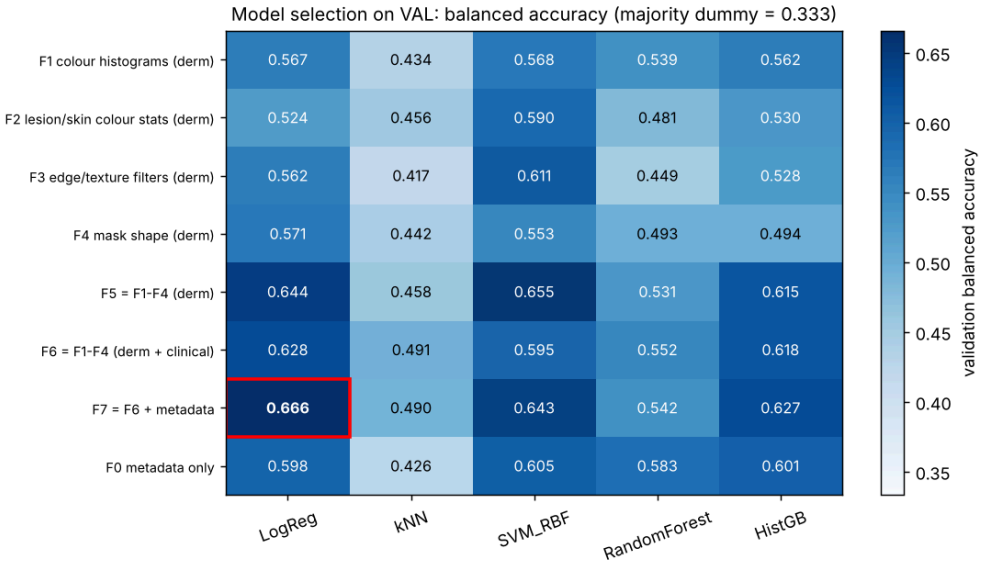


Figure 1. Validation balanced accuracy for every feature set × model (chance = majority dummy = 0.333). Red = selected.

- Every feature family beats chance. **Texture (0.61) and colour (0.59) are the strongest; shape is the weakest.** Combining them helps (F5: 0.655).
- The clinical view adds little (its masks are poor), but **metadata alone already reaches ≈ 0.60**. Images + metadata is best: **F7 + Logistic Regression (C = 0.01): val 0.666, CV 0.663 ± 0.052**.
- Linear models and the SVM beat trees and kNN. Random Forest and kNN predict almost no "Indeterminate" (88 train lesions). The top four configurations are within one CV standard deviation of each other: only 18 val lesions are Indeterminate.

3. Final results on the test set (749 lesions, used once)

Metric	Selected: F7 + LogReg	95 % CI	Majority dummy
Balanced accuracy (primary)	**0.669**	0.580 – 0.744	0.333
Macro-F1	0.550	0.508 – 0.594	0.273
Macro ROC-AUC (one-vs-rest)	0.820	0.768 – 0.859	0.500
Recall Benign / Indeterminate / Malignant	0.695 / 0.588 / 0.724	Malignant: 0.686 – 0.765	0 / 0 / 1
Precision Benign / Indeterminate / Malignant	0.644 / 0.115 / 0.870		– / – / 0.693
"Needs work-up" sensitivity / benign specificity	0.847 / 0.695	sens.: 0.817 – 0.878	1.000 / 0.000

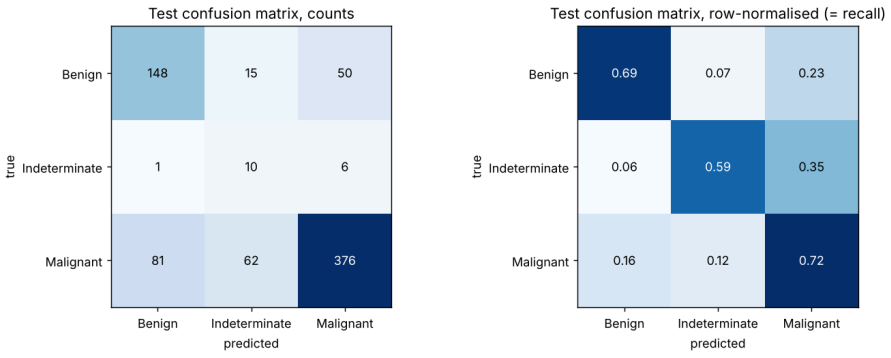


Figure 2. Test confusion matrix (left: counts, right: recall per true class).

- Test (0.669) ≈ val (0.666) ≈ CV (0.663): **no overfitting to the validation set** despite choosing among 40 candidates.
- **Melanoma is the main failure: 60 % of melanomas are predicted Benign**, while 96 % of nevi are correct. Global colour/texture statistics cannot see the local patterns that separate melanoma from nevus. 81 malignant lesions (16 %) are called benign. Indeterminate precision is only 0.12.
- **Which features are used** (grouped permutation importance on val): metadata (−0.118 balanced accuracy when shuffled) > dermoscopic colour (−0.109) > dermoscopic histograms (−0.083) > dermoscopic texture (−0.048) > clinical colour/texture (−0.03) > shape (≈ 0.01).
- **Age shortcut:** malignant recall is 0.43 for patients under 50, and benign recall is 0.41 for patients 70 and over. Women and men have the same balanced accuracy (0.66 / 0.67).
- Design check: early fusion of the two views beats averaging per-image predictions (val 0.666 vs 0.627). **Stretch, 8 classes:** balanced accuracy 0.416 (chance 0.125), macro AUC 0.862.

4. Conclusion

The best classical configuration is **class-balanced Logistic Regression on colour + texture + shape features of both views plus age/sex/site**. Its **test balanced accuracy is 0.669 [0.58, 0.74]** (macro AUC 0.820), twice the chance level. Colour is the most useful image cue, followed by texture. However, the images add only ≈ +0.06 over metadata alone, the model leans on age, and it misses most melanomas. This is the baseline the CNN in the next milestone must beat (same splits and metrics, reported with and without metadata), above all on melanoma recall.